UIDAI Data Hackathon 2026

Team Name: Encoders

Project title:- "Operational Risk Detection Framework for Aadhaar Biometric Activity Using Multi-Dimensional Analysis"

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Dataset Used: - 1.Biometric
2.Demographic
3.Enrollment

1.Problem:

During the past few years, Aadhaar enrolments and updates of biometrics have witnessed periodic surges in states and districts. These surges bring along multiple challenges such as system congestion, enrolment centre overload, authentication delays, and underutilization of infrastructure, among others. Therefore, developing an approach to reduce this operational stress during peak periods and enhancing system readiness has become one of the priorities at UIDAI for offering seamless services. In this hackathon, we are aiming to analyse anonymized Aadhaar biometric data with an objective to bring out the temporal, geographic, and demographic patterns behind surges and come up with data-driven and actionable frameworks that can be used to support UIDAI's proactive planning, risk mitigation, and policy decision-making.

UIDAI currently lacks early warnings and prioritized risks for districts, leading to the delay in responding to the surge in enrolments and biometric updates. Our analysis keeps the gaps covered by providing an explainable, data-driven method for risk scoring that makes proactive planning possible.

2.Objective:

For designing an explainable risk detection system, based on the temporal, geographic, and demographic biometric activity patterns to locate the structurally vulnerable states and districts, to enable proactive operational decisions by the UIDAI.

3.Approach:

Temporal Spike Detection:

Monthly biometric activity was collected at the national level to identify recurring seasonal and anomalous demand spikes.

Geographic concentration analysis:

State and district-level data was analysed for peak months to see if spikes occurred throughout the region or if they were limited to specific districts.

This helps us to identify hotspots contributing disproportionately to national-scale bursts of biometric activity.

Demographic attribution:

Biometric activity and demographic update were broken down for different age groups to identify whether national surges were child-centric or adult-centric update behaviour. This attribution gives a view of demand-side drivers associated with policy, welfare, and compliance.

District Risk Classification:

Districts were classified based on the magnitude of growth, absolute biometric load, as well as the concentration patterns within the states.

It helped us to identify top states and districts which contributed to the surges and the districts that displayed abnormally high spikes compared to the normal behaviour. This categorized structurally stable areas and those that are liable for sudden operational distress.

Policy-ready risk scoring:

A composite risk score was calculated using the time, geographical and demographic variables derive from Exploratory Data Analysis (EDA). This risk score divides the state and district into three categories into high, medium and low, district with high-risk score is at high probability of operational failures due to load, sudden spikes and concentration.

It enables district-level prioritization to be involved in proactive planning, monitoring, and intervention strategies.

4.Datasets Used:

Dataset	Source	Key Column	Dataset link
Biometric Dataset	UIDAI	date, state, district, pin code, bio_age_5_17, bio_age_17+	**cleaned_biometric.xls**
Demographic Dataset	UIDAI	date,state, district,pincode, demo_age_5_17, demo_age_17+	**cleaned_demographic.xls**
Enrolment Dataset	UIDAI	date,state, district,pincode, age_0_5,age_5_17, age_17+	**cleaned_enrollement.xls**

Table No.1

Biometric Dataset:

Biometric update data served as the main analytical foundation because it directly signals operational stress and risk in the Aadhaar system, due to its capacity demands, infrastructure requirements, and sharp surges. This makes it the most relevant indicator for detecting systemic load and informing policy interventions.

Analytically, the data was aggregated monthly and by region to reveal demand patterns and load concentrations, split by age group to identify demand drivers, and used at the district level to construct a risk scoring framework.

Demographic Dataset:

Demographic updates are easy and steady, making them a good way to track normal Aadhaar activity. When we compared these to biometric updates, we saw that big spikes in biometric activity didn't match increases in demographic updates. This shows that biometric surges are special events, not just part of general updates.

Enrolment Dataset:

We checked if more people signing up for Aadhaar led to more biometric activity and found there wasn't much of a link. Most of the system stress comes from how people use Aadhaar later, not from new enrolments. So, enrolment numbers help give context but aren't the main way to measure risk.

5.Methodology:

5.1 Data Cleaning

To ensure reliable and consistent analysis, all three datasets were cleaned as follows:

1. We made state and district names consistent, fixing any spelling or naming differences (For example Odisha and Orissa, Bangalore and Bengaluru).
2. When districts were split, merged, or renamed, we used the same names throughout to keep things clear.
3. Duplicate records were removed to avoid double counting.
4. Missing values were handled to keep the data accurate.

5.2 Feature Engineering

To make the raw data more useful for analysis, several key steps were taken:

1. We grouped the daily data into months so it would be easier to spot big trends without getting distracted by daily ups and downs.
2. We added up all the biometric updates to see how much pressure the system was under.

3. By checking how the numbers changed from month to month, we could quickly notice any sudden jumps.
4. We also looked at what portion of the activity came from each district to see which places were busiest.
5. Finally, we separated the data for adults and children to figure out which age group was using the system more.

5.3 Analytical Design

Univariate:

First, we looked at each dataset on its own to spot basic trends like the big spike in biometric updates in July 2025 and which states contributed most. We did the same for monthly enrolment numbers to see how they changed through the year.

Bivariate:

We compared different pairs of data mainly checking if a rise in enrolments led to more biometric updates, and which districts were growing fastest in biometric activity. We found that while enrolment sometimes leads to more biometric updates, most system stress comes from how the system is used, not just from new sign-ups.

Trivariate:

To apply the district risk model, a combination of growth rate, age groups, and location was considered. In the high-risk district, the growth rate was high, the main users are adults, and the districts are a few. For the medium-risk districts, the growth rate is steady, and the users are a combination of all age groups in several locations. The low-risk districts had slow growth rate and consistent growth. Thus, the growth of the system may result in different risks either in terms of locations, users, or time.

Tools used for analysis:

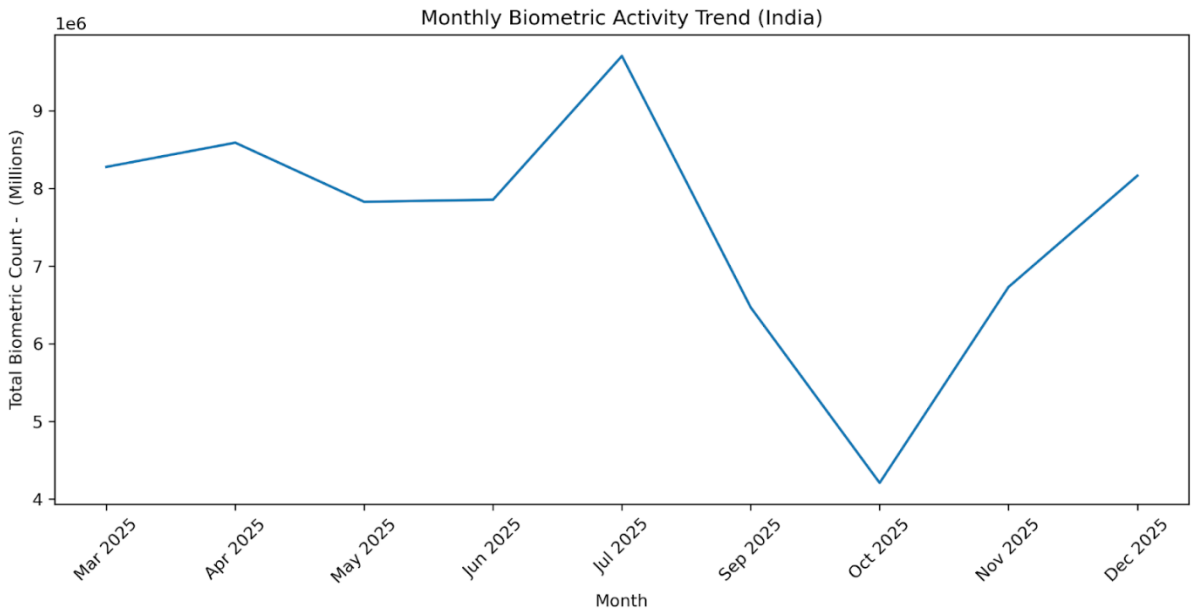
- **Python** for data processing and analysis
- **Pandas** for data manipulation and aggregation
- **Matplotlib and Seaborn** for visualisation and exploratory analysis

6.Data Analysis & Visualisation:

We have performed EDA (Exploratory data analysis) on our datasets to extract meaningful insights specifically through five types of stages of EDA .

Demographic and enrolment datasets were used as contextual baselines to distinguish routine Aadhaar activity from operationally intensive biometric surges.

EDA-1: Time-based trend analysis - Monthly Trend (India)



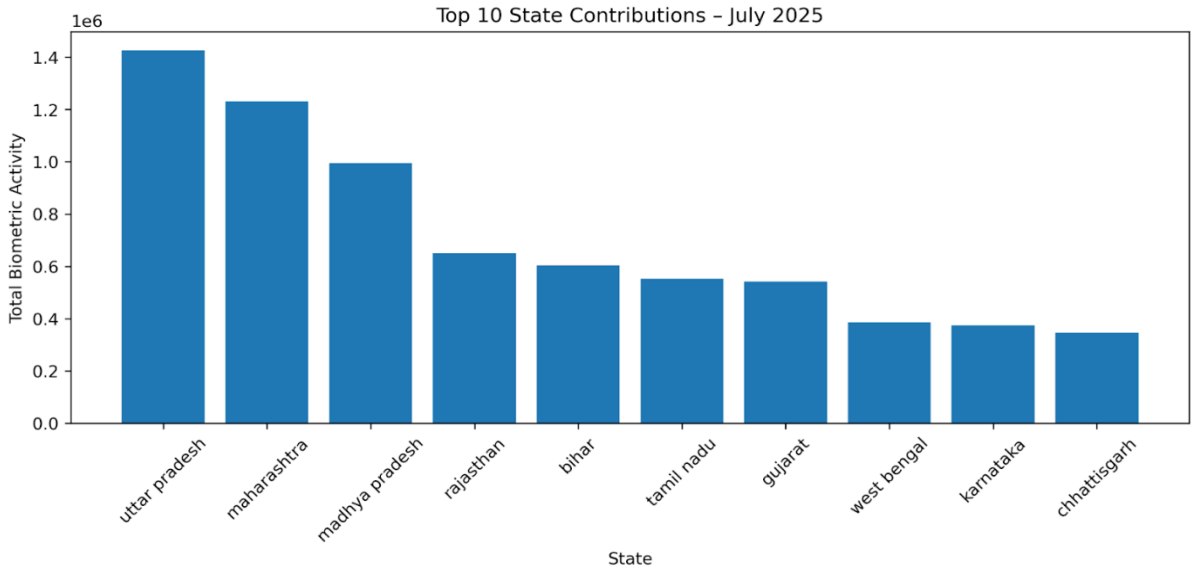
The accompanying line chart shows the no. of people (in millions) that updated their biometrics along the whole course of year. A **sharp** and Observable spike can be seen in the month of July which indicates a mass enrolment or update activity. Followed by a sudden drop (50%) from July to October and strong rebound after October which may indicate backlog accumulation from low-activity months.

UIDAI Recommendation: UIDAI should implement month-ahead biometric demand forecasting using historical enrolment/update data to:

1. Increase operator availability & allocate enrolment kits proactively to reduce workload and stress during peak months.
2. It directly affects UIDAI Infrastructure & Technology Division and Aadhaar Enrolment Ecosystem Management.

This will in turn reduce authentication failures, Faster enrolment and Improve citizen experience during peak months.

EDA-2: State Contribution (July)

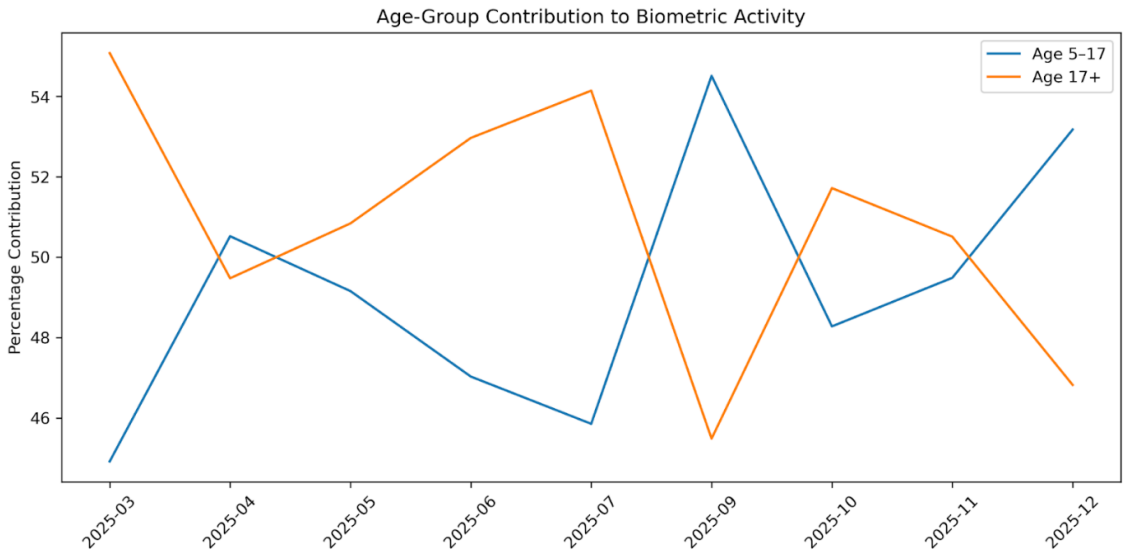


The above bar graph shows the top 10 states that contributed to the surge in July, and it is evident that the spike is not evenly distributed across India. A small set of large states drive most of the activity, mostly populous states.

This suggests targeted operational triggers or capacity-driven enrolment dynamics in high-volume states.

UIDAI Recommendation: UIDAI should do state weighed planning and pre-allocate resources to high volume states before known peak months. It will help UIDAI in better uptime during peak enrolment months and less repeat visits (especially critical in rural UP, Bihar, MP).

EDA-3: Age Group Contribution



The line graph suggests that the spike is primarily driven by 17+ age groups and not minors, the minor age group is concentrated in November month only. So, the July spike fits into adult-dominant biometric activity, indicating demand-side triggers such as welfare authentication, employment-linked verification, or compliance drives to update Aadhaar.

UIDAI Recommendation: Route adult update traffic to off-peak centers and reduce congestion during demand surges. This can also help UIDAI in policy coordination as policy benefits are one of the main causes for Aadhaar loads.

EDA 4: District Analysis

state	district	total_bio
maharashtra	nashik	83938
maharashtra	pune	83000
karnataka	bengaluru	76283
maharashtra	thane	74661
gujarat	ahmedabad	66780
maharashtra	jalgaon	62477
maharashtra	mumbai	52979
rajasthan	jaipur	52457
maharashtra	chhatrapati sambhajinagar	49607
maharashtra	nanded	46666
maharashtra	amravati	46551
maharashtra	solapur	45141
west bengal	south twenty four parganas	44924
uttar pradesh	prayagraj	44580
maharashtra	ahilyanagar	43881

Table 2: High absolute volume districts

The above Table 2 depicts that Urban and large districts and states dominate the overall spike due to their biometric volume these districts are always heavy contributors, even in normal months. It suggests that these districts require permanent infrastructure strengthening and capacity planning.

state	district	growth_pct
manipur	imphal east	1303.415560
manipur	thoubal	951.781857
manipur	bishnupur	903.983657
manipur	kakching	773.333333
manipur	imphal west	757.594937
manipur	senapati	330.225080
madhya pradesh	maihar	302.597403
manipur	ukhrul	302.295918
manipur	jiribam	253.956835
uttar pradesh	hamirpur	219.136582

Table 3: High growth districts

The above depicts those districts that displayed abnormal behaviour and extreme growth spikes compared to June 2025. Imphal east shows a growth percentage of 1300% compared to its previous month These are not large districts but displayed unusual growth rate due to something which contributed to their share of spike in July. In short, these areas are vulnerable to short-term overload and require early warning mechanisms.

Table 2 shows districts that are always contributors to the surges due to their high population whereas Table 3 shows districts that displayed anomaly compared to their normal behaviour. National biometric spikes are driven by small, localized surges rather than only large urban canters.

EDA-5: Structural Risk Typology

state	growth_pct
manipur	667.734513
ladakh	92.330383
uttar pradesh	66.500400
haryana	61.748512
dadra and nagar haveli and daman and diu	61.040609
rajasthan	56.017231
west bengal	44.690188
andaman and nicobar islands	42.405765
uttarakhand	38.861100
assam	32.966357

Table 4: State Growth Percentage

state	district	district_share_pct
lakshadweep	lakshadweep	100.000000
ladakh	leh	61.963190
puducherry	pondicherry	61.142343
goa	north goa	57.270270
dadra and nagar haveli and daman and diu	daman	51.930654
andaman and nicobar islands	south andaman	51.693266
meghalaya	west garo hills	47.583565
sikkim	east sikkim	42.260870
tripura	west tripura	32.676819
mizoram	aizawl	31.172650
manipur	imphal east	25.575766
delhi	north west delhi	23.945725

Table 5: State Growth Percentage

From the above tables we have classified States into 2 types –

Type1 - Scale States (Stable + Distributed), high volume states and activity spread across many districts. Like Maharashtra, UP, they are not seen in the above tables because EDA-5 surfaces extremes, not balanced states.

Type2 - Concentration Risk States (Spiky + Fragile) High growth OR sudden spikes and Activity dominated by 1–2 districts example - Manipur, Ladakh etc.

UIDAI Recommendation EDA-5 illustrates that the state's total growth rate can create a wrong perception of development at the state level if it is predominantly powered by a single district. UIDAI can introduce a District Concentration Risk Index (DCRI) which can prevents service disruption during surges. UIDAI can create a Growth Classification Dashboard to examine the overall growth of states if the load is evenly spread among districts.

7.Solution Framework:

Each entity (State or District) is scored on 5 independent risk signals derived directly from our EDAs.

Total Risk Score=R1+R2+R3+R4+R5

1.R1 (July Spike Exposure) - From EDA-1 + EDA-2

Condition	Score
State in Top 10 contributors in July	+20 points
Otherwise	0 points

2.R2 (Growth Volatility) - From EDA-4(Captures unpredictability and surge risk)

Growth rate	**Score**
≥ 40%	+25
25–39%	+15
<25%	0

3.R3 — District Concentration Risk (Single-district dominance = fragility)

Top District Share	**Score**
≥ 60%	+25
40–59%	+15
<40%	0

4.R4 — Absolute Load Pressure (Chronic infrastructure stress)

District Volume Rank	Score
Top 10 nationally	+20
Top 25	+10
otherwise	0

5.R5 — Adult Skew Indicator (Adult updates correlate with policy/program rollouts.)

Adult Share (17+)	Score
≥ 55%	+10
Otherwise	0

6. Risk Classification Thresholds

Score Range	Risk level	UIDAI Meaning
≥ 60	High risk	Immediate monitoring
40-59	Medium risk	Watchlist + early prep
<40	Low risk	Routine operations

state	district	risk_score	risk_level	R1	R2	R3	R4	R5
rajasthan	jaipur	75	High	20	25	0	20	10
west bengal	murshidabad	65	High	20	25	0	10	10
ladakh	leh	60	High	0	25	25	0	10
west bengal	west medinipur	55	Medium	20	25	0	0	10
rajasthan	sikar	55	Medium	20	25	0	0	10
rajasthan	jhunjhunu	55	Medium	20	25	0	0	10
rajasthan	ganganagar	55	Medium	20	25	0	0	10
rajasthan	dhaulpur	55	Medium	20	25	0	0	10
rajasthan	chittaurgarh	55	Medium	20	25	0	0	10
rajasthan	alwar	55	Medium	20	25	0	10	0
west bengal	alipurduar	55	Medium	20	25	0	0	10
west bengal	cooch behar	55	Medium	20	25	0	0	10
west bengal	darjeeling	55	Medium	20	25	0	0	10
west bengal	east midnapore	55	Medium	20	25	0	0	10
uttar pradesh	prayagraj	55	Medium	20	25	0	10	0

Table 6: Risk Scores

Our Risk score solves the exact cause of UIDAI which is identifying districts/states that need attention first. It tells us that in the observed spike, these districts pose the highest operational risk.

This risk score is scalable across datasets and reproducible, it uses only the UIDAI data. It converts EDA insights - policy driven - execution.

- **Why Rajasthan dominates**: Because Rajasthan districts satisfy multiple risk conditions simultaneously: like it is in top 10 states in surges(R1), Some districts rank high nationally(R4), Rajasthan shows strong July growth(R2) etc. So, Rajasthan districts accumulate medium high-risk scores, even without extreme concentration.
- **Why Manipur showed growth but didn't have a high-risk score**: It showed explosive growth only for a short period of time. And it was relative change. Moreover, it didn't satisfy multiple risk signals like not been in top 10 July state**.**

- **Why High-Volume states like Maharashtra, UP are missing:** Our risk model Identifies where failure probability is high not where load is high, these states fall into type1 - which are stable, distributed and predictable.
- **Why is R3 (district concentration) zero for most districts:** Because district-level concentration is rare, it triggers only when 1 district dominates 40-60% of the state activity. Large states like Rajasthan or UP naturally spread activity.

Validation of our Risk Score:

risk_level	avg_growth_pct	avg_adult_share_pct	district_count
High	64.35	64.69	3
Medium	55.64	45.84	155
Low	27.96	57.96	627

Table 7: Risk Score

The Findings from the Table-7 validate that higher-risk districts consistently show higher growth and higher adult biometric dominance than medium and low districts. This proves that risk Score reflects real stress and not arbitrary weights.

8.Impact & Applicability:

1. **Key Insights**: Biometric load has predictable seasonal spikes, mostly from adults
   UIDAI Action: Focus on actual biometric usage trends and seasonal patterns for planning, instead of relying on enrolment data.
   Expected Impact:
   - No waiting in long queues
   - Fewer repeat visits
   - Better experience for people, without needing permanent extra infrastructure.

2. **Key Insights**: System risk is focused on just a few districts. Most biometric stress comes from a handful of fast-growing, adult-heavy districts. Most other areas are stable.
   UIDAI Action: Use a district-level risk system to catch trouble early and respond quickly where it's needed most.
   Expected Impact:
   - Prevents crises in high-risk areas
   - Pre allocate resources to high-risk districts
   - Keeps the system strong without extra burden

9.Applicability:

The proposed framework is scalable across states and districts, allowing UIDAI to prioritise risk dynamically as demand patterns evolve. It does not require any additional data beyond existing Aadhaar enrolment and update records and can be deployed immediately as a decision-support tool.

10.Limitations:

This analysis is based on anonymised Aadhaar datasets and does not incorporate external policy calendars or programme-level timelines, which may influence biometric activity patterns. Additionally, detailed data on enrolment centre capacity and staffing levels was not available, limiting direct assessment of on-ground operational readiness.

11.Jupyter Notebooks & Dataset Used:

Drive Link: Link